SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Coding challenge

Code of Conduct:

I D.Uday Kiran(192425161) certify that ~~this submission is my original work and that I have adhered to the guidelines specified for this assessment.~~
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

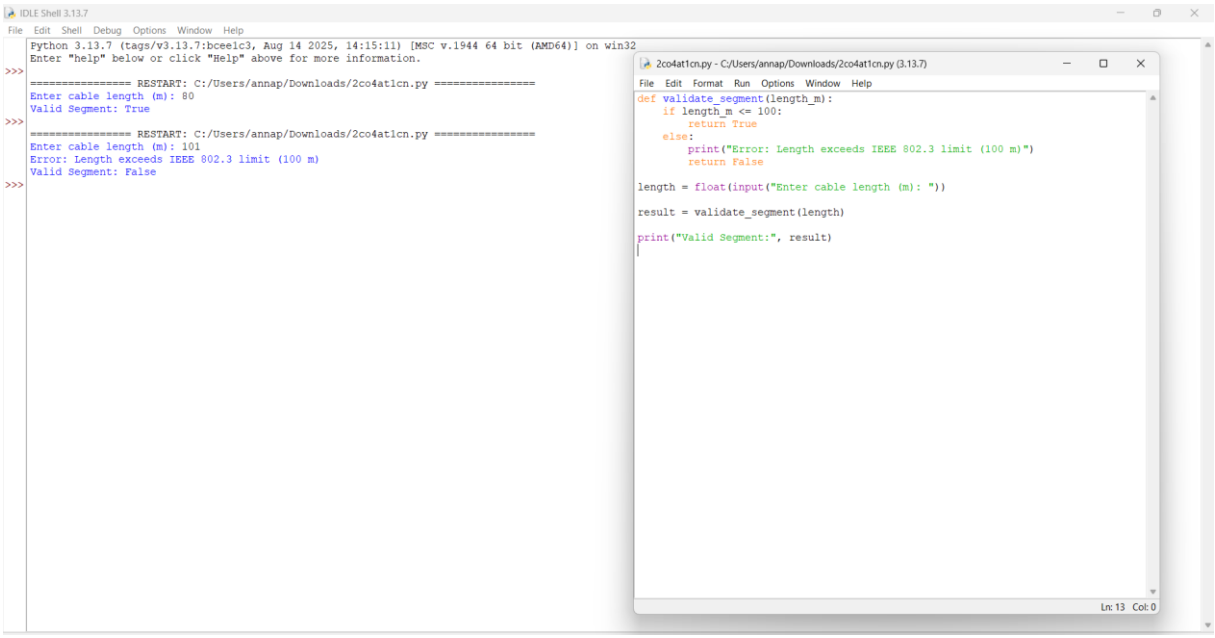
D.Uday Kiran

Annexure A

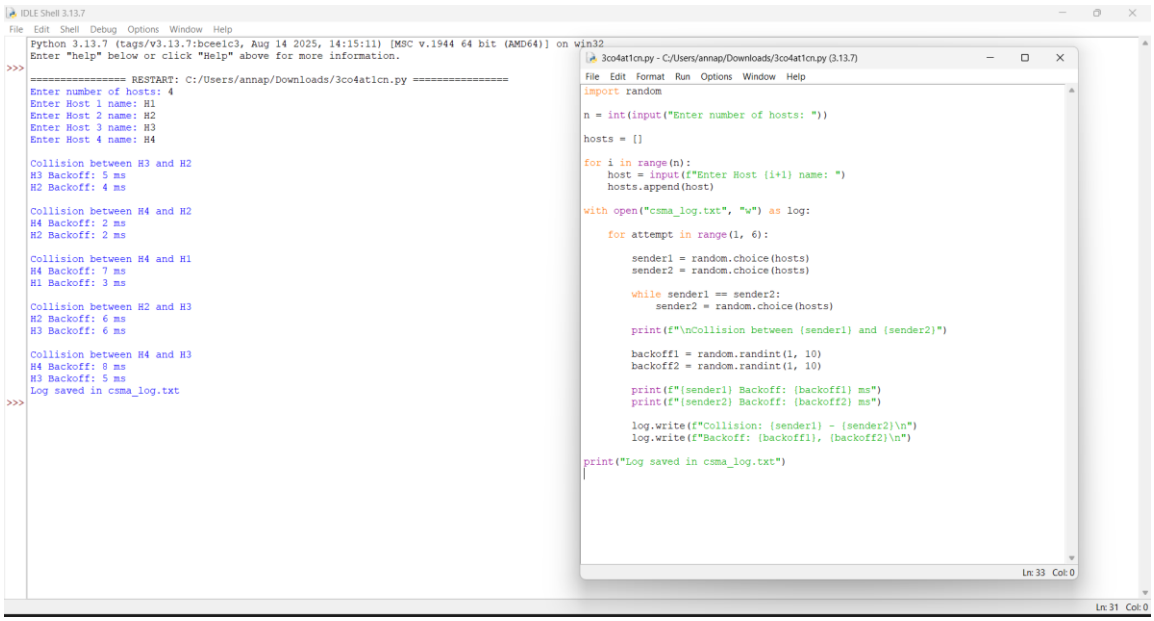
Coding challenge

1. Write a Python script that generates a star topology diagram (ASCII or graphical) given n hosts and a central switch. The script must print each host label, the switch name, and the connectingCat6 UTP cable length.

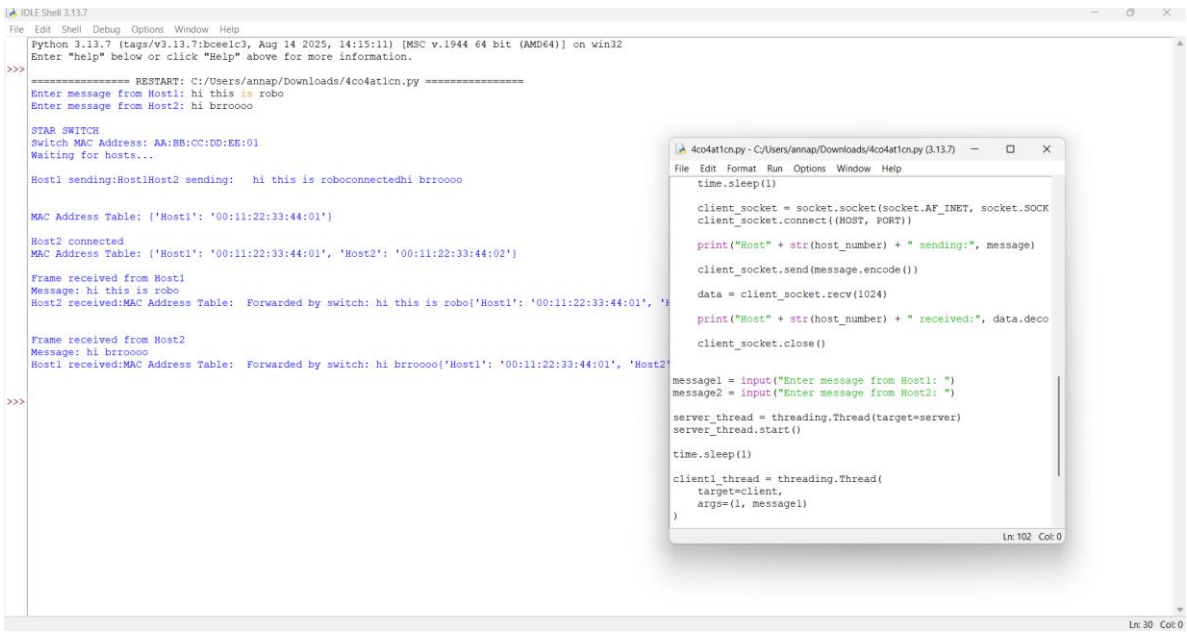
2. Code a function `validate_segment(length_m)` that returns True if the UTP segment is within IEEE 802.3 limits (≤100 m) and False with an error message otherwise. Include at least 3 test cases.



3. Write a script that simulates CSMA/CD on a star topology with 4 hosts. Log every collision detection event and the back-off interval to a file named `csma_log.txt`.



4. Using Python's `socket` library, write a client-server program that models two hosts communicating through a star switch. Demonstrate message forwarding by printing the switch's MAC address table after each frame transmission.



5. Create a class `StarSwitch` with methods `add_port(host_id)`, `remove_port(host_id)`, and `broadcast(frame)`. Broadcast must send a frame to all ports except the source and log each delivery.